

PAPER_1559_M_MU_MUON_0_10566

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PAPER_1559 — Muon Mass = $m_\mu = F_{\text{TRZ}^2} \cdot SO_5 + F_{\text{TRZ}^2} \cdot (SSQ^2 + SSQ^3 + SSQ^4) \approx 0.10566 \text{ GeV}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Particle Physics

Date: June 18, 2026

Status: CLOSED — 0.040% closure from PAPER_1209HH_S660

Observation

PAPER_1209HH_S660 (Particle Physics Unified Proof Set) lists **Muon Mass** with the integer-primitive expression:

``

Muon Mass = $m_\mu = F_{\text{TRZ}^2} \cdot SO_5 + F_{\text{TRZ}^2} \cdot (SSQ^2 + SSQ^3 + SSQ^4) \approx 0.10566 \text{ GeV}$

``

Status: **0.040%**.

UQFF Closed Identity

``

$m_\mu = F_{\text{TRZ}^2} \cdot SO_5 + F_{\text{TRZ}^2} \cdot (SSQ^2 + SSQ^3 + SSQ^4) \approx 0.10566 \text{ GeV } 0.040\%$

``

Physical Interpretation

Notable structure: muon mass is **purely F_{TRZ^2} -suppressed** — uses only the $F_{\text{TRZ}^2} \times (SO_5 + SSQ\text{-polynomial})$ combination.

This demonstrates UQFF's natural separation between heavy/light fermions:

- Heavy fermions (top, b, c, τ) use A_5 , SO_5 , D_{crit} dominant terms
- Light fermions (μ , e) use only F_{TRZ^2} polynomial in SSQ

The F_{TRZ^2} prefactor universally suppresses light-fermion masses by a factor of 100 ($=SO_5^2$) — natural lepton hierarchy from integer primitives.

NOT REPLACEMENT

CODATA/PDG treats Muon Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S660
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_mu_muon_0_10566"})`

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